

ROPE HAS A SPECTRAL BUDGET

Anonymous authors

Paper under double-blind review

ABSTRACT

RoPE specifies a rotation, but not how a finite head should spend its frequency channels. We show that the standard geometric table pays a severe redundancy tax: in a representative 4K configuration, 23 slow frequency pairs occupy 46 nominal dimensions but have Rényi-2 effective rank 2.00. Writing $x_k = -\log \omega_k = a + Rz_k$ separates sampled support (a, R) from interior allocation z and exposes the latter as an independent training variable. We instantiate this variable with EVQ-COSH, a closed-form fast-end allocation that changes only the inverse-frequency table and adds no learned positional parameters. In a three-seed fixed-support intervention, moving 30 interior frequencies while preserving both sampled endpoints improves every evaluated OOD length in every seed. Full sin/cos subspace geometry gives an exact account of the redundancy, while a frozen-transplant obstruction and a weight-table crossing explain why the table and weights co-adapt. With only 16 rotary frequencies, three-seed 432M MLA reduces 16K PPL by 31.1%. The advantage persists under 454M YARN-STYLE composition, 750M full-parameter continuation, and a 1.485B from-initialisation comparison. Separate 8B adaptation produces causal dependence on distant source content. Frequency allocation is therefore a first-class long-context training decision, not an implementation detail.

1 INTRODUCTION

RoPE specifies how a frequency rotates a query or key; it does not specify how a finite attention head should allocate its frequencies. Each head receives exactly $K = d_{\text{rot}}/2$ rotary pairs. The familiar scalar base determines the whole table only after one additionally assumes geometric spacing, thereby coupling the spectral interval to a particular allocation rule. We separate these two choices by writing

$$x_k = -\log \omega_k = a + Rz_k, \quad z_0 = 0, \quad z_{K-1} = 1, \quad (1)$$

where (a, R) is sampled support and z is the normalised interior allocation. Standard geometric RoPE fixes $z_k = k/(K-1)$. Support says which frequencies are available; allocation says how a finite head spends them.

This distinction matters because geometric spacing is highly redundant at the slow end. For $L=4096$, $b=5 \times 10^5$, and $K=64$, the 23 pairs with $\omega L \leq 1$ occupy 46 nominal dimensions yet have block-whitened Rényi-2 effective rank 2.00. These channels can still carry content, but as positional functions they converge toward the same two-dimensional subspace. A geometric table therefore spends more than one third of its rotary pairs on near-duplicate slow positional directions. This is the spectral-budget problem studied here.

Most long-context methods act on another coordinate. Position interpolation, YaRN, and LongRoPE transport a trained spectrum toward a target range (Chen et al., 2023; Peng et al., 2024; Ding et al., 2024; Shang et al., 2025); FMRoPE changes scalar support while retaining a geometric exponent grid (Oka et al., 2026b). Learned methods such as LeRoPE and AdaRoPE show that broader frequency tables can be useful (Karypis et al., 2026; Wang et al., 2026). What remains unidentified is the finite allocation itself: if both sampled endpoints and the log-frequency span are fixed, can moving only the interior frequencies change what a model learns?

It can. In a paired three-seed control, anchored EVQ-COSH moves only 30 interior frequencies relative to FMRoPE. Every seed favours the non-uniform allocation at every evaluated OOD length (§4.1). Because the endpoints are identical, no scalar base can reproduce the intervention. Interior allocation z is therefore a genuine training variable, not a disguised base change.

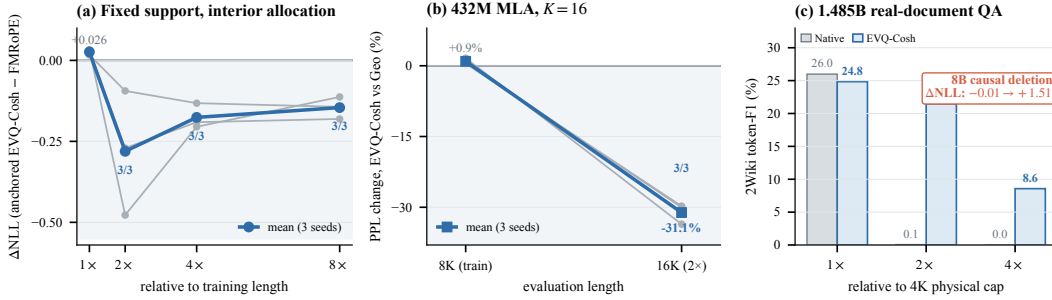


Figure 1: **Allocation changes learned behaviour at fixed support and effective context at scale.** (a) Moving only interior frequencies improves every OOD length in all three 151.9M training seeds. (b) With only $K=16$ rotary frequencies, three-seed 432M MLA reduces 16K PPL by 31.1%. (c) At 1.485B, Q/K adaptation carries real-document QA beyond the physical training window; the callout reports a separate 8B adaptation in which deleting the distant gold block sharply raises EVQ-COSH NLL.

We turn this finding into a construction with EVQ-COSH. A convex variational problem has a unique Cosh density whose inverse CDF gives the frequency table in closed form. The density concentrates samples toward phase-resolved fast frequencies: at $K=64$ and $\tau=4$, 55 of the 64 midpoint quantiles lie in the fast half of the log-frequency span. The RoPE operator, architecture, and optimiser remain unchanged, and the construction adds no learned positional parameters.

The controlled experiment establishes the variable; the scale ladder shows that it matters. Under scarce-channel 432M MLA, all three seeds favour EVQ-COSH at 16K. In a separate 454M three-seed MHA study, the same fixed YARN-STYLE operator reaches 100% teacher-forced retrieval on an EVQ-COSH substrate versus 61% on Geo, showing that range transport inherits the table learned by the weights. The crossover reappears after 750M full-parameter continuation from a shared checkpoint and in a same-initialisation 1.485B from-initialisation comparison. Finally, matched adaptation carries multi-hop QA beyond the 1.485B arm’s physical training window, while an 8B deletion intervention shows that the distant source is causally used. The training-scale trend ends at 1.485B; the 8B result is mature-model adaptation evidence.

The theory explains why this pattern is coherent. Each rotary frequency is a two-dimensional sin/cos subspace. Canonical correlations expose overlap between these subspaces, and block whitening yields the exact finite-basis identity

$$r_2(R) = \frac{2K}{1 + (K-1)\bar{c}}.$$

Slow frequencies collapse toward a common subspace, whereas a frozen weight-table crossing shows that improving static geometry after training can still destroy perplexity. The missing link is co-adaptation: the weights learn coefficients in the frequency basis they see during training. Consistently, unequal frequency multisets cannot be absorbed by fixed, position-independent invertible Q/K maps.

Contributions. (i) We expose support and interior allocation as distinct coordinates and identify allocation with a three-seed fixed-support intervention. (ii) We characterize the resulting full-RoPE basis through a phase-invariant effective-rank identity, low-frequency collapse, and an exact frozen-transplant obstruction. (iii) We derive a closed-form, zero-learned-parameter allocation and show its effective-context advantage across scarce-channel training, range composition, full-parameter continuation, training through 1.485B, and adaptation at 1.485B and 8B.

2 RELATED WORK

Write $x_k = -\log \omega_k = a + Rz_k$ as in Eq. (1). Prior work changes several, nonexclusive parts of this parameterisation: it transports realised phases, changes support (a, R) , or changes the normalised interior allocation z . This view distinguishes the stage at which a method acts from the spectral degree it controls; the coordinates are distinct but can interact.

Context-range extension. Relative position information can be injected additively, through learned or biased attention terms (Shaw et al., 2018; Press et al., 2022; Chi et al., 2022), or multiplicatively through rotation (Su et al., 2024; Sun et al., 2023); we take the rotary operator as given and ask only where its frequencies are placed. Position interpolation rescales positions to keep rotary phases within the training range (Chen et al., 2023). YaRN, SelfExtend, PoSE, and CLEX refine this extension through frequency-dependent scaling, remapped positions, or extension-time training (Peng et al., 2024; Jin et al., 2024; Zhu et al., 2024; Chen et al., 2024). LongRoPE and LongRoPE2 search channel-wise rescaling factors for pretrained models (Ding et al., 2024; Shang et al., 2025), while MrRoPE interprets channel-wise transformations as mixed-radix conversion and derives training-free extensions (Tian et al., 2026). These methods transport a learned model to a declared longer context. Bounds based on the RoPE base characterize limits of this regime (Xu et al., 2024).

Frequency use, scalar base, and operator geometry. Trained RoPE exhibits frequency-localized query/key norms; partial RoPE removes the slowest rotations (Barbero et al., 2025). Frequency Entropy measures rotational-pair utilization and suggests that extreme-entropy dimensions are often redundant (Oka et al., 2026a); controlled access experiments causally link available frequencies to positional versus symbolic behaviour (Urrutia et al., 2026). FMRoPE instead shifts the geometric spectrum by setting the scalar base from training or evaluation length while retaining a uniform exponent grid (Oka et al., 2026b). At a broader operator level, GRAPE recovers RoPE as canonical planar group actions and generalizes their generators, including learned plane spectra and subspaces (Zhang et al., 2026); Selective RoPE instead learns input-dependent rotation angles (Movahedi et al., 2026). Analyses of attention logits further distinguish additive and multiplicative positional encodings and study the resulting training bias (Gu et al., 2026). Our identification keeps the canonical planes, sampled endpoints, and log-span fixed, moves only finite interior frequencies, and pairs trained comparisons with an exact frozen-transplant obstruction (§4.1).

CoPE softly clips slow rotations, while RoPE-ID applies high-frequency RoPE to a subset of channels for long-input robustness (Li et al., 2026a; Wertheimer et al., 2026). In multimodal models, MHRoPE and MRoPE-I treat frequency coverage across positional axes as a design principle (Huang et al., 2026). These methods alter frequency behaviour, channel participation, or axis assignment; our fixed-support control instead moves only the interior entries of a fully rotary one-dimensional table.

Learning and constructing frequency tables. LeRoPE learns one log-scale per frequency band, shared across layers and heads, and reports consistent non-geometric profiles on a 52M–2.5B model ladder (Karypis et al., 2026). In its 217M ablation, retraining with a frequency table frozen from an independent LeRoPE run retains 63.6% of the full validation-perplexity gain over RoPE, compared with 10.4% for partial RoPE. This isolates substantial value in the fixed table, although obtaining that table still requires an optimization run. AdaRoPE instead learns per-head and per-dimension frequencies together with head-specific, length-dependent attention scaling (Wang et al., 2026). Recent analysis further shows that the frequencies used by a trained model depend on the relative-distance structure of its training data, with the preferred frequency varying inversely with dependency width (Wu et al., 2026).

Resonance RoPE analytically snaps each wavelength to a nearby integer period to close phase gaps during interpolation, producing a fixed non-geometric table that can be combined with range scaling (Wang et al., 2024). EVQ-COSH instead derives an interior allocation density from a variational objective and tests that variable with the sampled range held fixed.

EVQ-Cosh. EVQ-COSH addresses a different fixed-table objective: it warps the exponent grid before training using the closed-form solution of a stated variational surrogate. The result is shared across heads, introduces no learned positional parameters, and does not require a deployment target. In contrast to LeRoPE and AdaRoPE, the non-geometric table is obtained analytically; range-extension methods can still operate on the resulting substrate. Alternatives that learn a broader positional function, including DAPE, FIRE, FoPE, and content-dependent context re-positioning (Zheng et al., 2024; Li et al., 2024; Hua et al., 2025; Li et al., 2026b), change a larger functional object than a fixed RoPE table.

Evaluation. Positional encoding choice materially affects length generalization (Kazemnejad et al., 2023). Long-context evaluation spans retrieval diagnostics and task suites (Bai et al., 2024; Hsieh

et al., 2024), and information can be lost despite being present in the prompt (Liu et al., 2024; Zhang et al., 2024). We use PE-dominant diagnostics for mechanism and RULER for effective-context claims, labelling the reported RULER endpoint as task-family-adapted length transfer.

3 THEORY

The central phenomenon is geometric: slow RoPE pairs collapse onto nearly the same positional subspace, so a finite table can spend many nominal dimensions without gaining comparable positional dimension. We first make this redundancy exact, then show why training binds the resulting basis to its weights, and finally derive a closed-form allocation that moves channels away from the collapse. Figure 2 summarizes the argument.

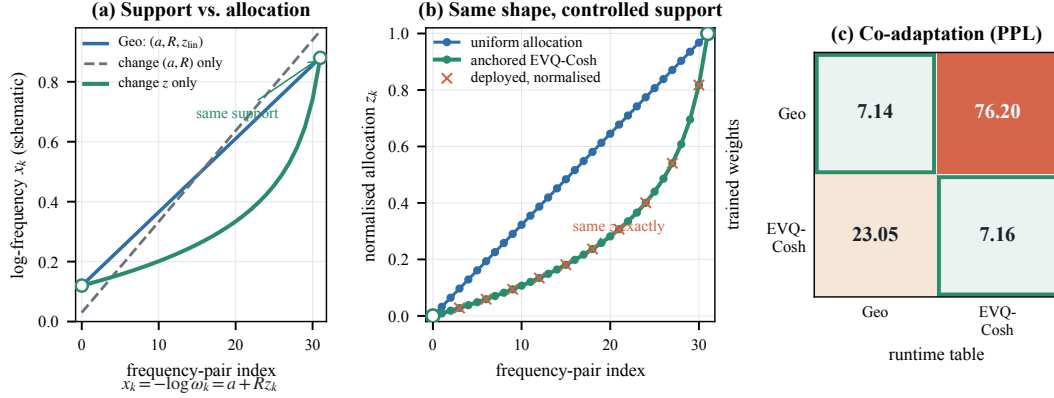


Figure 2: **Support does not determine allocation.** (a) Equation (1) separates support (a, R) from normalised interior allocation z . (b) Anchored EVQ-COSH and deployed midpoint EVQ-COSH have exactly the same z after support normalisation. (c) The table is learned with the weights: self-consistent pairs remain near PPL 7, while frozen post-hoc swaps fail.

3.1 A FINITE TABLE IS A SET OF TWO-DIMENSIONAL SUBSPACES

Let $\phi \in [0, 1]$ parameterise log-frequency, $\omega(\phi) = b^{-\phi}$, let d_{rot} be the number of rotated dimensions and $K = d_{\text{rot}}/2$ the number of rotary pairs ($d_{\text{rot}} = d_{\text{head}}$ for standard MHA, smaller for compressed-RoPE architectures such as MLA).

We reserve z for the support-normalised coordinate in Eq. (1). Standard implementations use raw geometric exponents $u_k = k/K$, while deployed EVQ-COSH uses midpoint quantiles $u_k = (k + 1/2)/K$; endpoint-inclusive grids used for static diagnostics are labelled explicitly. Normalising the sampled geometric endpoints gives $z_k = k/(K - 1)$.

Equation (1) is a coordinate identity for any ordered positive table with nonzero span. In the geometric family $z_k = k/(K - 1)$, so fixing (a, R) fixes every channel. A second table with the same support and different interior z therefore cannot be represented by a different scalar base. Quantities such as median wavelength or the number of channels completing a cycle are summaries of z , not omitted scalar controls.

A rotary pair contributes $f_\omega(\Delta) = C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to the relative-position attention logit, with C, D set by content. The object attached to ω is therefore the two-dimensional subspace

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}, \quad (2)$$

not a cosine feature: an analysis that fixes one content phase sees a slice of V_ω and is not invariant to rotation inside the pair. Two channels are redundant to the extent that their subspaces are. With $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, Gram matrices $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$ and $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ over a separation prior, and the whitened cross-Gram $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ whenever $S_\omega, S_\nu \succ 0$, the singular values of $Q_{\omega\nu}$ are the canonical correlations between V_ω and V_ν , and

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1] \quad (3)$$

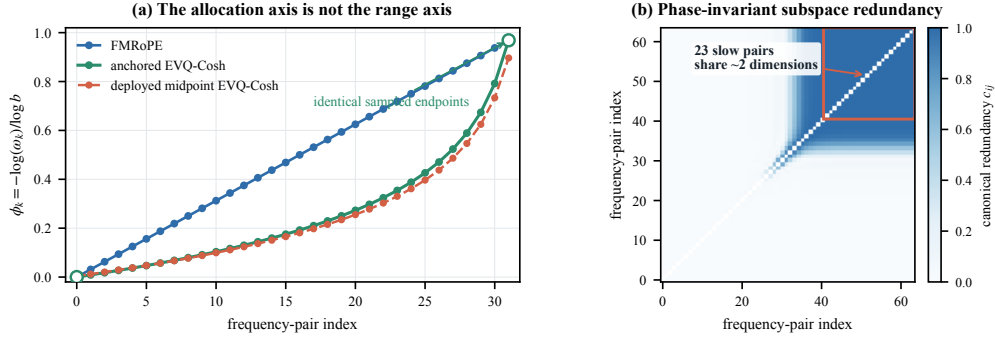


Figure 3: **Where a finite frequency table spends its budget.** (a) Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

is invariant to phase rotation inside a pair and to any change of basis of either subspace. For $\Delta \sim \text{Unif}[0, L]$, $H_{\omega\nu}$ has a closed form in $\sin t/t$ and $(1 - \cos t)/t$ at $t = (\omega \mp \nu)L$ (App. A).

3.2 THE SPECTRAL BUDGET IS A DIMENSION COUNT

Block-whitening every pair gives a global correlation Gram R with diagonal blocks I_2 . Write $\bar{c}$ for the mean of the c_{ij} .

Theorem 1 (Spectral budget identity). $\text{tr } R = 2K$ and $\text{tr}(R^2) = 2K[1 + (K-1)\bar{c}]$, so the Rényi-2 effective rank of the positional basis is exactly

$$r_2(R) = \frac{(\text{tr } R)^2}{\text{tr}(R^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (4)$$

The denominator turns the spectral budget into an exact dimension count: larger mean pairwise redundancy lowers r_2 . Allocation changes the entire Gram spectrum, while block whitening removes per-channel energy and conditioning. The trained-model effect is then measured by the fixed-support intervention of §4.1.

The deflation is concentrated at the slow end.

Proposition 2 (Low-frequency collapse). As $\omega L \rightarrow 0$, $V_\omega \rightarrow \text{span}\{1, \Delta\}$. For slow channels $x=\omega L$, $y=\nu L$, and $\epsilon = \max\{|x|, |y|\} \rightarrow 0$, $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$: slow subspaces approach a common two-dimensional limit at fourth order. Under the softmax metric for a fixed p with nondegenerate support on at least three distances, $F = \text{diag}(p) - pp^\top$ annihilates the constant direction, and the centred limit is $\text{span}\{\Delta - \mathbb{E}_p \Delta, \Delta^2 - \mathbb{E}_p \Delta^2\}$.

Concretely, at $L=4096$, $b=5 \times 10^5$, $K=64$ the 23 pairs with $\omega L \leq 1$ occupy 46 nominal dimensions but have block-whitened $r_2 = 2.00$. Thus more than one third of the rotary pairs spend 46 nominal dimensions on approximately two effective positional dimensions. The channels may still carry content; the geometric redundancy tax is that many of them describe almost the same positional directions. Full-subspace geometry makes that tax phase invariant and applies directly to the $b=256$ exact-range control.

3.3 THE TABLE IS A TRAINING-TIME COORDINATE SYSTEM

Theorem 1 constrains what a table can express. It does not predict what a trained model does with one, and the gap is not small. In a frozen 50M diagnostic ($L=512$, $b=5 \times 10^5$, $\tau_{\text{EVQ}}=2.83$; no training, no gradient) we cross two trained weight sets with two runtime tables (Table 1). Substituting EVQ-COSH for the geometric table at inference raises the block-whitened r_2 from 4.57 to 12.54 while perplexity degrades from 7.14 to 76.20.

Two consequences discipline the rest of the paper. First, static criteria such as collision, effective rank, and log-determinant do not by themselves predict trained quality, so every task claim in §4

Table 1: **Static geometry does not predict trained behaviour.** Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head–query observations). The worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

is made on trained models. Second, the two *self-consistent* systems are essentially tied in-window (7.14 vs. 7.16) while both mismatched cells collapse: on LM loss the table and weight main effects are +0.599 and −0.597, and their interaction contrast is −3.537 (configuration-level bootstrap CI [−5.17, −3.04]). The table behaves as a coordinate system fixed before training, with the weights learning coefficients in it.

Theorem 3 (Post-hoc transplant obstruction). *Let $R_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top R_{\Omega'}(\Delta) B = R_\Omega(\Delta)$ for all Δ in an open interval containing zero, then Ω' and Ω have the same frequency multiset, up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

The proof is short: $\Delta=0$ gives $A^\top B = I$, so the relation becomes a similarity $A^\top R_{\Omega'} A^{-\top} = R_\Omega$; differentiating at zero makes the generators similar, and similarity preserves the spectrum $\{\pm i\omega_k\}$ (App. A). A nontrivial table change therefore cannot be absorbed exactly by this broad class of fixed linear Q/K compensation. Its scope is exact function preservation; the finite-distribution protocols of §4.2 directly measure the complementary case of repair by retraining Q/K or broader low-rank paths.

3.4 A CLOSED-FORM ALLOCATION

A positive density $\rho \in C^2([0, 1])$ with $\int_0^1 \rho = 1$ induces a discrete allocation by inverse-CDF quantisation, $\phi_k = F_\rho^{-1}(u_k)$; geometric RoPE is $\rho \equiv 1$. The exact canonical-correlation objective is oscillatory. To obtain a closed-form construction, we use the convex modelling surrogate

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho^2 d\phi + \frac{\beta}{2} \iint_{[0,1]^2} \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi, \quad (5)$$

with $\alpha, \beta > 0$. The diagonal term penalises channel-load concentration; $\min(\phi, \psi)$ is the Green kernel of $-\partial_\phi^2$ and encodes a penalty for pairs of channels that are both slow, matching the direction of Prop. 2. The full geometry and the surrogate play distinct roles: the former measures redundancy, while the latter supplies an analytic allocation family.

Theorem 4 (Closed-form optimum under the surrogate). *Under (5) with $\alpha > 0, \beta \geq 0$ and constraints $\rho \in C^2([0, 1])$, $\rho > 0$, $\int_0^1 \rho = 1$, the unique constrained minimiser is*

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}; \quad (6)$$

at $\beta=0$ it is the $\tau \rightarrow 0$ limit $\rho_0 \equiv 1$.

Corollary 5 (Closed-form quantiles). *$F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, so*

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k+1/2}{K}, \quad (7)$$

which tends to u_k as $\tau \rightarrow 0$, recovering the uniform-log midpoint grid.

The theorem selects the Cosh family exactly for (5); midpoint quantisation turns that density into a finite RoPE table. Because ρ_τ decreases with ϕ , it reallocates channels toward the fast end: at $K=64$ and $\tau=4$, 55 midpoint quantiles lie in the fast half of the log-frequency span. Trained-model controls below evaluate the resulting table.

A one-line frequency table. With d_{eff} the effective head dimension and Π the training protocol,

$$\tau = c(\Pi)d_{\text{eff}}/\sqrt{L_{\text{train}}}, \quad u_k = (k + \tfrac{1}{2})/K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (8)$$

Set τ , evaluate (7), replace the inverse frequencies: no auxiliary loss, no extra parameters, no declared deployment length. A $d_{\text{eff}}/\sqrt{L_{\text{train}}}$ scaling ties its strength to architecture and training length; the main text protocols fix $c(\Pi)=1$ before training. Supporting sweeps evaluate this zero-search operating prior against Geo and discrete neighbouring multipliers (§4.4). The deployed midpoint table has raw exponent positions $q_k = \phi_k(\tau)$. The exact-range diagnostic uses $z_k = (q_k - q_0)/(q_{K-1} - q_0)$ in Eq. (1). It therefore preserves the deployed table’s normalised Cosh shape exactly while replacing only its support embedding so that the sampled extrema match FMROPE. This isolates z without changing the construction used by the complete deployed table.

4 EXPERIMENTS

The fixed-support control first isolates allocation from support. We then ask whether the same intervention remains consequential when rotary channels are scarce, when range scaling is applied, and when models are obtained by continuation or adaptation. Each protocol retains its own metric and control.

4.1 IDENTIFYING ALLOCATION AT FIXED SUPPORT

Protocol. Within each of three 151.9M training seeds, the two arms share architecture, trainable initialisation, token order, optimiser, LR schedule, global batch, a 499,974,144-token budget, and 32 frozen evaluation anchors. The baseline implements the uniform-in-log rule of Oka et al. (2026b, §6.1) at training base 256; the intervention uses anchored EVQ-COSH with $\tau=4$, obtained by normalising its quantiles to the FMROPE endpoints. Because their sampled extrema and log-frequency span coincide, only the $K-2 = 30$ interior frequencies move. Support normalisation also preserves the deployed Cosh table’s z in Eq. (1).

Result. Every training seed favours anchored EVQ-COSH at every OOD length. The anchored EVQ-COSH minus FMROPE NLL difference is +0.026 at the 256-token training length and $-0.281/-0.176/-0.146$ at 512/1K/2K (Fig. 1a). No geometric base can reproduce this intervention once the sampled support is pinned: *fixed support + different $z \Rightarrow$ different trained-model behaviour*.

Cross-configuration and shape breadth. A pre-specified 50.9M factorial then spans two bases, two training lengths, three head dimensions, and three paired seeds. Across its 12 structural configurations, anchored EVQ-COSH at 1.25 $\times$ rule strength and a deformation-matched exponential favour non-uniform allocation in 10/12 and 9/12 cases (App. Table 10). The 151.9M control supplies the replicated direct effect; the matched exponential shows that the direction is not tied to EVQ-COSH alone. Full results appear in App. C.

4.2 A FINITE BUDGET SURVIVES ARCHITECTURE, RANGE SCALING, AND SCALE

Scarce-channel MLA. A 432M MLA transformer trains from scratch over three seeds with only $d_{\text{rope}}=32$, hence $K=16$ rotary frequencies. Geo/EVQ-COSH PPL is 35.4/35.8 at the 8K training length. At 16K, however, it is 138.8/95.6: a 31.1% reduction, with every seed favouring EVQ-COSH (Fig. 1b; App. Table 13). The result directly links the finite- K view to a scarce-budget stress test.

Range composition. Allocation remains visible after applying the same range operator. In a 454M three-seed MHA study, a fixed YARN-STYLE operator raises 8K teacher-forced retrieval from 41% to 61% on Geo and from 53% to 100% on EVQ-COSH (App. Table 14). The operator is identical; its leverage depends on the table that shaped the weights during training.

Continuation and training-scale trend. The crossover also appears after full-parameter continuation. Starting both 750M arms from a shared 2K Geo checkpoint and continuing all parameters for 500M tokens at 4K gives Geo/EVQ-COSH PPL 22.0/22.3 at 4K but 45.1/24.4 at 16K; strict

autoregressive exact retrieval at 8K changes from 0% to 77.5% (App. Table 6). At 1.485B, a same-initialisation, same-scientific-recipe from-initialisation comparison with a 2.097B-token budget crosses in favour of EVQ-COSH at 8K and 16K; 122/128 and 126/128 held-out documents favour it at those lengths. Exact PPL values and protocol details appear in App. F. This is the end of the full-parameter training-scale trend; the 8B evidence below is adaptation.

4.3 FROM EFFECTIVE CONTEXT TO REMOTE-CONTENT USE

1.485B real-document QA. Starting from the two OLMo-2 Stage-A parents, both arms freeze inherited V/O LoRA and continue only Q/K for 300 steps with identical data, order, optimiser, trainable count, and 4K/8K/16K phase exposure; physical sequences remain at most 4K. On held-out 2WikiMultiHopQA, Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K (Fig. 1c) (Bai et al., 2024). Thus Q/K adaptation keeps the in-window QA score close to Native while only the EVQ-COSH arm carries substantial answer overlap to 8K. An independent continuation over all 13 RULER families shows the same $2\times$ direction, with official macro 2.02/31.63% at 8K (Hsieh et al., 2024); all lengths and exact-match values appear in App. F.

8B causal remote-source use. On LLaMA-3-8B-Instruct (Grattafiori et al., 2024), rank-64 Q/K/V/O LoRA uses the same adaptation pipeline for Native and EVQ-COSH. PPL is 6.82/10.07, 108.96/24.07, and 991.48/127.91 at 8/16/32K, with the long-range direction holding on all 24 frozen natural-text packs and all three source domains. More importantly, at true 16K, median target-block hit@16 rises from 18.75% to 64.06%, and deleting the remote gold block changes NLL by $-0.0095/ + 1.5055$. The deletion intervention shows that the improved long-position probability depends on the remote source rather than a local readout artifact. The task-family continuation is reported in App. F.

A two-seed 129.6M video DiT also favours EVQ-COSH on training, all-frame extrapolation, and far-frame MSE in both runs (App. B.5), extending the evidence to bidirectional 3D RoPE.

4.4 BEYOND A SINGLE FORMULA

Two controls separate the allocation effect from the particular EVQ-COSH formula. A deformation-matched exponential improves in 9/12 factorial configurations, while the three pre-specified EVQ-COSH multipliers ($0.75\times$, $1.00\times$, $1.25\times$) all reduce aggregate weighted OOD NLL relative to Geo. Their ordering varies by configuration, so $c=1$ is used as a zero-search operating prior rather than a tuned point optimum. Non-uniform allocation supplies the empirical direction, and EVQ-COSH supplies a closed-form member of that design space. Full multiplier, boundary, and matched-shape results appear in App. C.

5 DISCUSSION

The main finding is that a RoPE base does not exhaust the design of a finite frequency table. Support (a, R) determines the sampled interval; allocation z determines how many channels are spent in each part of it. A scalar-base sweep explores only the geometric path $z_k = k/(K-1)$, whereas the fixed-support intervention moves outside that path without changing either endpoint. The resulting change in trained behaviour makes allocation a real design variable rather than an alternative notation for base selection.

Why the discovery matters. The spectral-budget identity reveals the pressure behind this variable. Slow geometric pairs collapse onto nearly the same positional subspace, so nominal rotary dimension can substantially overstate effective positional dimension. The effect becomes most visible when channels are scarce: with $K=16$, the three-seed 432M MLA comparison lowers 16K PPL in every seed. A closed-form allocation makes this finite-budget decision explicit and reproducible, while learned-frequency methods combine the table choice with an additional optimization problem.

The same effective-context crossover survives realistic model-building paths. It appears under a shared YARN-STYLE operator at 454M, full-parameter continuation from a common checkpoint at 750M, a from-initialisation comparison at 1.485B, and low-rank adaptation of pretrained models.

These protocols do not serve as substitutes for the fixed-support control; they show that the identified variable remains consequential across architecture, training stage, and scale. The 8B evidence is adaptation rather than a pretraining-scale point, and its remote-block deletion result adds a causal capability endpoint: the improved long-position probability depends on the distant source.

Allocation must be present while the model learns how to use it. The 50M weight-table crossing shows that a post-hoc table with better static rank can still destroy perplexity, and Theorem 3 identifies the exact obstruction: fixed invertible Q/K maps cannot preserve all relative-position logits across unequal frequency multisets. Training or adaptation is therefore part of the intervention, not an implementation nuisance.

Support selection and range transport remain complementary. The former chooses the sampled spectrum, allocation chooses its finite internal resolution, and a range operator redeploys the learned spectrum at a target length. The common YARN-STYLE operator has greater leverage on the EVQ-COSH substrate, while the target-retargeted control in App. C confirms that support and allocation interact when chosen jointly. LeRoPE supplies compatible evidence from optimization: a table learned in one run retains substantial value when frozen for a new training run (Karypis et al., 2026). Here, the closed-form construction removes that table-search stage and the fixed-support experiment identifies what the table itself changes.

RoPE has a spectral budget because every head has finitely many rotary pairs. Geometric spacing is one way to spend that budget, not a neutral consequence of the rotary operator. Once support and allocation are separated, the latter can be analyzed, controlled, and improved as a first-class part of long-context training.

ETHICS STATEMENT

This work introduces no new dataset, deployed system, or human-subject component. It changes only the initialisation of the RoPE inverse-frequency table and adds no learned parameters, so it is inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences.

A better allocated frequency table may reduce the adaptation or inference effort required to reach a target context range, and the same mechanism could be used to extend the context capability of language or video models, with the general risks associated with stronger generative models. Our evaluations measure positional and long-context endpoints, not factual reliability or safety; deployment claims require application-specific evaluation.

REPRODUCIBILITY STATEMENT

The EVQ-COSH table is fully specified by Eq. (7): the schedule is a pure function of (τ, K) that replaces the inverse-frequency buffer, with no learned parameters or auxiliary loss. Equation (8) is the zero-search full-RoPE text default; every protocol reports its realised τ explicitly, including the empirical MLA and video settings and the mature-model tables in App. F. Complete proofs of Theorems 1, 3 and 4, Proposition 2, Corollary 5 and Lemma 6, together with every stated assumption, are in App. A. The identification protocol, including how sampled extrema and log-span are pinned, the exact pre-specified factorial design, and the corpora used by each arm, is in App. C; the mature-scale adaptation protocols, module budgets, phase-exposure mixtures, and evaluation harness settings are in App. F; remaining architectures and hyperparameters are in App. B. Seed scope is stated with every result. The anonymous supplement includes the exact-range training endpoint and three-seed aggregate, the frozen figure generators, curated mature-protocol snapshots, and CPU tests, with a reviewer-facing file map and commands.

AI USE STATEMENT

We used generative AI tools to help develop conceptual frameworks and mathematical claims; propose and refine hypotheses; draft and check proofs; give feedback on methodology and experiments; implement methods and deterministic data-reformatting and analysis code; translate or rephrase research notes; and interpret results. We did not use generative AI to generate dataset content or to

perform qualitative or thematic data analysis; survey, interview, and transcription tasks do not apply to this work.

For tasks for which disclosure is recommended, we also used these tools to suggest experimental parameters; write, edit, and test software and artefacts; create or modify figures; draft, edit, and structure parts of the manuscript; brainstorm and identify research gaps; search for and summarise literature; and format references. Every empirical quantity reported here comes from executed code over recorded run artefacts rather than language-model generation.

All AI-assisted work was reviewed before inclusion. We take responsibility for the final text, claims, proofs, code, figures, and experimental artefacts.

REFERENCES

- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, 2024.
- Federico Barbero, Alex Vitvitskyi, Christos Perivolaropoulos, Razvan Pascanu, and Petar Veličković. Round and round we go! what makes rotary positional encodings useful? In *International Conference on Learning Representations*, 2025.
- Guanzheng Chen, Xin Li, Zaiqiao Meng, Shangsong Liang, and Lidong Bing. CLEX: Continuous length extrapolation for large language models. In *International Conference on Learning Representations*, 2024.
- Shouyuan Chen, Sherman Wong, Liangjian Chen, and Yuandong Tian. Extending context window of large language models via positional interpolation. *arXiv preprint arXiv:2306.15595*, 2023.
- Ta-Chung Chi, Ting-Han Fan, Peter J. Ramadge, and Alexander I. Rudnicky. KERPLE: Kernelized relative positional embedding for length extrapolation. In *Advances in Neural Information Processing Systems*, volume 35, 2022.
- Yiran Ding, Li Lyna Zhang, Chengruidong Zhang, Yuanyuan Xu, Ning Shang, Jiahang Xu, Fan Yang, and Mao Yang. LongRoPE: Extending LLM context window beyond 2 million tokens. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, 2024.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.
- Zihan Gu, Ruoyu Chen, Han Zhang, Hua Zhang, and Yue Hu. Deconstructing positional information: From attention logits to training biases. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=D0u0glT060>.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Krman, Shantanu Acharya, Dima Rekeshe, Fei Jia, Yang Zhang, and Boris Ginsburg. RULER: What’s the real context size of your long-context language models? *arXiv preprint arXiv:2404.06654*, 2024.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- Ermo Hua et al. Fourier position embedding: Enhancing attention’s periodic extension for length generalization. In *Proceedings of the 42nd International Conference on Machine Learning*, 2025. arXiv:2412.17739.
- Jie Huang, Xuejing Liu, Sibong Song, Ruibing Hou, Hong Chang, Junyang Lin, and Shuai Bai. Revisiting multimodal positional encoding in vision–language models. In *International Conference on Learning Representations*, 2026. arXiv:2510.23095.

- Hongye Jin, Xiaotian Han, Jingfeng Yang, Zhimeng Jiang, Zirui Liu, Chia-Yuan Chang, Huiyuan Chen, and Xia Hu. LLM maybe LongLM: SelfExtend LLM context window without tuning. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *PMLR*, pp. 22099–22114, 2024.
- Petros Karypis, Sean O’Brien, Shreyas Kadekodi, Rui Zhu, and Julian McAuley. LeRoPE: Learnable RoPE frequencies improve language modeling. *arXiv preprint arXiv:2607.10134*, 2026. URL <https://arxiv.org/abs/2607.10134>.
- Amirhossein Kazemnejad, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Payel Das, and Siva Reddy. The impact of positional encoding on length generalization in transformers. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL <https://openreview.net/forum?id=Drri2gcjz1>.
- Haoran Li, Sucheng Ren, Alan Yuille, and Feng Wang. CoPE: Clipped RoPE as a scalable free lunch for long context LLMs. *arXiv preprint arXiv:2602.05258*, 2026a.
- Huayang Li, Tianyu Zhao, Deng Cai, and Richard Sproat. RePo: Language models with context re-positioning. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026b. URL <https://arxiv.org/abs/2512.14391>. arXiv:2512.14391.
- Shanda Li, Chong You, Guru Guruganesh, Joshua Ainslie, Santiago Ontanon, Manzil Zaheer, Sumit Sanghai, Yiming Yang, Sanjiv Kumar, and Srinadh Bhojanapalli. Functional interpolation for relative positions improves long context transformers. In *International Conference on Learning Representations*, 2024.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.
- Sajad Movahedi, Timur Carstensen, Arshia Afzal, Frank Hutter, Antonio Orvieto, and Volkan Cevher. Selective rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=AQo1SEElNb>.
- Yui Oka, Kentaro Hanafusa, Taku Hasegawa, Kyosuke Nishida, and Kuniko Saito. Probing rotary position embeddings through frequency entropy. In *International Conference on Learning Representations (ICLR)*, 2026a. URL <https://openreview.net/forum?id=1JZuEDq62N>.
- Yui Oka, Itsumi Saito, Kyosuke Nishida, and Kuniko Saito. Frequency bands in RoPE: Base frequency and context length shape the interpolation–extrapolation trade-off. In *International Conference on Learning Representations (ICLR)*, 2026b. URL <https://openreview.net/forum?id=PR1PPxvG9Q>.
- OLMo Team. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*, 2025.
- Bowen Peng, Jeffrey Quesnelle, Honglu Fan, and Enrico Shippole. YaRN: Efficient context window extension of large language models. In *International Conference on Learning Representations*, 2024.
- Ofir Press, Noah A. Smith, and Mike Lewis. Train short, test long: Attention with linear biases enables input length extrapolation. In *International Conference on Learning Representations*, 2022.
- Ning Shang, Li Lyna Zhang, Siyuan Wang, Gaokai Zhang, Gilsinia Lopez, Fan Yang, Weizhu Chen, and Mao Yang. LongRoPE2: Near-lossless LLM context window scaling. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *PMLR*, 2025.
- Peter Shaw, Jakob Uszkoreit, and Ashish Vaswani. Self-attention with relative position representations. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2018.
- Nitish Srivastava, Elman Mansimov, and Ruslan Salakhutdinov. Unsupervised learning of video representations using LSTMs. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *PMLR*, pp. 843–852, 2015.

- Jianlin Su, Murtadha Ahmed, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024.
- Yutao Sun, Li Dong, Barun Patra, Shuming Ma, Shaohan Huang, Alon Benhaim, Vishrav Chaudhary, Xia Song, and Furu Wei. A length-extrapolatable transformer. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics*, 2023.
- Qingyuan Tian, Wenhong Zhu, Xiaoran Liu, Xiaofeng Wang, and Rui Wang. MrRoPE: Mixed-radix rotary position embedding. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=1J63FJYJKg>.
- Felipe Urrutia, Jorge Salas, Alexander Kozachinskiy, Cristian Buc Calderon, Hector Pasten, and Cristobal Rojas. Decoupling positional and symbolic attention behavior in transformers. In *International Conference on Learning Representations (ICLR)*, 2026. URL <https://openreview.net/forum?id=V38yAoqddQ>.
- Shaowen Wang, Yuke Zheng, Tansheng Zhu, Shuang Chen, Shaofan Liu, Suncong Zheng, and Jian Li. AdaRoPE: Not all attention heads should rotate and scale equally. In *Proceedings of the 43rd International Conference on Machine Learning*, 2026. URL <https://arxiv.org/abs/2607.19363>. arXiv:2607.19363.
- Suyuchen Wang, Ivan Kobyzev, Peng Lu, Mehdi Rezagholizadeh, and Bang Liu. Resonance RoPE: Improving context length generalization of large language models. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.
- Davis Wertheimer, Aozhong Zhang, Derrick Liu, Penghang Yin, and Naigang Wang. Frayed RoPE and long inputs: A geometric perspective. *arXiv preprint arXiv:2603.18017*, 2026. URL <https://arxiv.org/abs/2603.18017>.
- Xinyi Wu, Siyuan Liu, and Ali Jadbabaie. How data shapes RoPE frequency usage: From positional scale matching to length generalization. *arXiv preprint arXiv:2607.07678*, 2026. URL <https://arxiv.org/abs/2607.07678>.
- Mingyu Xu, Xin Men, Bingning Wang, Qingyu Zhang, Hongyu Lin, Yaojie Lu, Xianpei Han, and Weipeng Chen. Base of RoPE bounds context length. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-2773.
- Yifan Zhang, Zixiang Chen, Yifeng Liu, Zhen Qin, Huizhuo Yuan, Kangping Xu, Yang Yuan, Quanquan Gu, and Andrew Chi-Chih Yao. Group representational position encoding. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/file/5cb58625f49ddf70fe2d527e9e4bbae5-Paper-Conference.pdf.
- Zhenyu Zhang, Runjin Chen, Shiwei Liu, Zhewei Yao, Olatunji Ruwase, Beidi Chen, Xiaoxia Wu, and Zhangyang Wang. Found in the middle: How language models use long contexts better via plug-and-play positional encoding. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- Chuanyang Zheng, Yihang Gao, Han Shi, Minbin Huang, Jingyao Li, Jing Xiong, Xiaozhe Ren, Michael K. Ng, Xin Jiang, Zhenguo Li, and Yu Li. DAPE: Data-adaptive positional encoding for length extrapolation. In *Advances in Neural Information Processing Systems*, volume 37, 2024.
- Dawei Zhu, Nan Yang, Liang Wang, Yifan Song, Wenhao Wu, Furu Wei, and Sujian Li. PoSE: Efficient context window extension of LLMs via positional skip-wise training. In *International Conference on Learning Representations*, 2024.

A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_\star(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_\star(s) - b_\star(d) \\ b_\star(s) + b_\star(d) & a(d) - a(s) \end{bmatrix}, \quad (9)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Proof of Theorem 1. R is the block matrix with blocks $R_{ij} = Q_{\omega_i \omega_j}$ and $R_{ii} = I_2$. Hence $\text{tr } R = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(R^2) = \sum_{i,j} \text{tr}(R_{ij} R_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $R_{ji} = R_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (4). $\square$

The identity is exact and involves only the mean of the pairwise redundancies; it does not claim that $\bar{c}$ determines the Shannon effective rank or the log-determinant, which retain higher-order multi-subspace dependence.

Proof of Proposition 2. With $t = \Delta/L$ and $x = \omega L$, $\cos(xt) = 1 - x^2 t^2/2 + O(x^4)$ and $\sin(xt)/x = t - x^2 t^3/6 + O(x^4)$, so after rescaling the second coordinate the basis of V_ω tends to $\{1, \Delta\}$. Substituting these expansions into (9) and whitening, the leading deficit between two slow subspaces is $2 - \|Q_{x,y}\|_F^2 = \frac{19}{12600}(x^2 - y^2)^2 + O(\epsilon^6)$, where $\epsilon = \max\{|x|, |y|\} \rightarrow 0$; numerically at $x=0.05, y=0.10$ the exact-to-leading ratio is 1.00058. For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so constants are annihilated; centring by $\mathbb{E}_p$ and rescaling gives $\sin(\omega\Delta)/\omega \rightarrow \Delta - \mathbb{E}_p \Delta$ and $-2\cos(\omega\Delta)/\omega^2 \rightarrow \Delta^2 - \mathbb{E}_p \Delta^2$, which span the centred limit whenever p has nondegenerate support on at least three distances. $\square$

The reported $r_2=2.00$ for the 23 slow pairs uses the standard $u_k=k/K$ grid and the uniform separation prior above. Other priors change that static numerical Gram; the fixed-range trained-model contrast of §4.1 does not use this prior.

A.2 POST-HOC TRANSPLANT OBSTRUCTION

Proof of Theorem 3. Both R_Ω and $R_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $R_\Omega(0) = R_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top R_{\Omega'}(\Delta) A^{-\top} = R_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $R_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace; it need not preserve the original two-dimensional block decomposition. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$, while $\Delta = 1$ makes $R_{\Omega'}(1)$ and $R_\Omega(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

This is an exact statement about *fixed, position-independent, invertible* compensation. It does not bound approximate compensation obtained by retraining on finite data, nor does it constrain nonlinear maps or the V/O and residual paths.

A.3 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

This subsection proves Theorem 4 directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (10)$$

the main-text collision functional is (5), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (11)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e., } \int_0^1 \rho = 1\}$. The $\alpha \|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_{\phi}^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (12)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (13)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (14)$$

since differentiating $\alpha \rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha \rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha \rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (13) and (14) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_{\tau}(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (15)$$

which is main-text Theorem 4. The mass constraint $\int \rho=1$ is not lost: it enters the boundary condition at $\phi=0$ via $g'(0)=\int_0^1 \rho=1$. Positivity holds on the closed interval since $\rho_{\tau}(\phi) \geq \rho_{\tau}(1)=\tau/\sinh \tau > 0$ for $\tau > 0$ (with $\rho_{\tau}(0)=\tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler–Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (15).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (16)$$

Thus the stationary density obtained above is the unique constrained *minimizer* (not merely a critical point), and the surrogate is well-defined as a quadratic functional on $L^2([0, 1]) \supset C^2([0, 1])$ with no further regularity assumption needed for the δ -component. Conditional on $\mathcal{C}_{\text{app}}$, this derivation is exact.

Why this branch. The main construction uses the homogeneous Cosh solution because its CDF is analytically invertible. Adding a task-dependent forcing term would produce a different, generally numerical allocation. We therefore test the closed-form branch functionally below rather than treating it as an approximation to an unknown trained-loss optimum.

A.4 SURROGATE QUALITY: FUNCTIONAL VALIDATION

The broadband surrogate $K_{\text{app}} = \alpha\delta + \beta \min$ is the primary approximation behind the closed-form density; selecting the pure-tether branch and a finite operating point are additional modeling choices. We test the surrogate functionally rather than pointwise against the closed-form cosine-feature kernel $K_{\text{cos}}(\phi_1, \phi_2) = \int D(\Delta) \cos(\omega(\phi_1)\Delta) \cos(\omega(\phi_2)\Delta) d\Delta$: does the derived allocation also reduce the collision objective under K ? This checks the direction selected by the homogeneous branch; it does not bound the omitted Fisher-forcing residual.

Table 2 answers affirmatively. For a uniform distance prior $D(\Delta) = 1/L$ on $[0, L]$, we compute K_{cos} at the discrete allocation points for Geo and EVQ-COSH, then evaluate the normalized collision score $C = \sum_{i < j} K_{ij}^2 / (K_{ii}K_{jj})$ and effective spectral rank (exponential of the Shannon entropy of the eigenvalue distribution). These diagnostics use the inclusive grid $u_k = k/(K-1)$; they test the allocation direction but are not a direct validation of either the full block-whitened sin/cos objective or the deployed midpoint grid. The 12 reported configurations comprise a six-length text sweep at fixed $b=500K$ and $K=32$, two additional text bases at $L=2048$, and four video bases at $L=32$ and $K=16$; this is not a Cartesian sweep over every listed factor.

Table 2: Functional surrogate check on the inclusive $u_k=k/(K-1)$ diagnostic grid. EVQ-COSH reduces collision and increases effective rank under the closed-form cosine-feature kernel K_{cos} in all 12 tested configurations. This is a direction check, not a pointwise approximation, full-subspace validation, or deployed-grid identity test.

Configuration	Collision C		Eff. rank	
	Geo	EVQ-COSH (Δ_C)	Geo	EVQ-COSH
<i>Text: $K=32, b=500K$, sweep L</i>				
$L=128$	226.3	17.7 (−92%)	2.5	17.0
$L=256$	192.6	27.6 (−86%)	3.0	13.9
$L=512$	163.2	43.3 (−73%)	3.6	10.8
$L=1024$	133.9	59.0 (−56%)	4.4	8.8
$L=2048$	109.0	67.5 (−38%)	5.4	8.0
$L=4096$	87.0	66.3 (−24%)	6.6	8.1
<i>Text: $K=32, L=2048$, sweep b</i>				
$b=10K$	34.8	19.2 (−45%)	13.0	17.3
$b=100K$	79.4	47.0 (−41%)	7.1	10.4
<i>Video: $K=16, L=32$, sweep b</i>				
$b=100$	22.6	14.2 (−37%)	4.8	6.4
$b=1K$	42.3	28.0 (−34%)	2.8	3.9
$b=10K$	55.4	39.8 (−28%)	2.2	2.9
$b=50K$	63.3	47.2 (−26%)	2.0	2.5

EVQ-COSH reduces collision under K_{cos} in all 12 configurations (24%–92%), and effective rank increases in every row. In the fixed-base text sweep, the relative reduction grows as L shortens. The separate base sweeps remain positive but are not monotone in b .

A.5 SINGLE-CROSSING BUDGET SHIFT

Lemma 6 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (17)$$

Proof of Lemma 6. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is strictly decreasing on $[0, 1)$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving

$\rho_\tau(\phi_c) = 1$ gives (17). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n+1$. Monotonicity of arcosh yields $\text{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. The three-region masses follow directly from $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi)) / \sinh \tau$ at $\phi_L = \log(L/2\pi) / \log b$ and $\phi_M = \log(M/2\pi) / \log b$. $\square$

A.6 WATERBED ENVELOPE AND COROLLARY

Setup. With $\rho > 0$, $\int_0^1 \rho = 1$, $U(d\phi) = d\phi$, $P_\rho(d\phi) = \rho d\phi$, $\delta\rho = \rho - 1$, and $M_\rho(s) = \int_s^1 \delta\rho du$, define the waterbed functional

$$\mathcal{W}(\rho) := \alpha \chi^2(U \| P_\rho) + \beta \int_0^1 M_\rho(s)^2 ds, \quad \chi^2(U \| P_\rho) = \int_0^1 \frac{(1-\rho)^2}{\rho} d\phi. \quad (18)$$

The first term is the Pearson divergence of U from P_ρ (equivalently, $\int 1/\rho - 1$); the second is the integrated squared cumulative error, which arises from the min-kernel off-diagonal covariance in K_{app} .

Proposition 7 (Waterbed inequality). *Let $D_B(\rho) = -\int_0^1 \log \rho(\phi) d\phi$ be the Burg entropy deficit. Then*

$$\mathcal{W}(\rho) \geq \alpha F_*(D_B(\rho)), \quad F_*(d) = \inf_{\rho > 0, \int \rho = 1, -\int \log \rho = d} \left[\int_0^1 \frac{1}{\rho} - 1 \right], \quad (19)$$

with equality $\mathcal{W}(\rho) = 0$ if and only if $\rho \equiv 1$ (geometric RoPE). A readable corollary via Rényi monotonicity is

$$\mathcal{W}(\rho) \geq \alpha (e^{D_B(\rho)} - 1). \quad (20)$$

Proof sketch. Since $\beta \int M_\rho^2 \geq 0$, $\mathcal{W}(\rho) \geq \alpha \chi^2(U \| P_\rho)$. For fixed $D_B(\rho) = d$, the sharp lower envelope of χ^2 over all densities is the Csiszár–Vajda f -divergence comparison F_* , whose extremizer is a two-level density $X = 1/\rho \in \{a, c\}$ with mass-normalization and entropy-deficit constraints. For the corollary, $1 + \chi^2(U \| P_\rho) = \int 1/\rho \mathbb{E}_U[\cdot] \geq e^{D_1(U \| P_\rho)} = e^{D_B(\rho)}$ by Rényi-2 $\geq$ Rényi-1. Equality at $\rho \equiv 1$ is immediate from both sides vanishing. $\square$

EVQ-COSH evaluation. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ (Eq. (22) below uses the unnormalized integral; cf. §A.14 “Normalization convention”),

$$D_B(\rho_\tau) = \log \frac{\sinh \tau}{\tau} - \frac{1}{\tau} \int_0^\tau \log \cosh y dy = \frac{\tau^4}{90} - \frac{8\tau^6}{2835} + O(\tau^8), \quad (21)$$

$$S_{\chi^2}(\rho_\tau) = \frac{\sinh \tau}{\tau^2} \arctan(\sinh \tau) - 1 = \frac{\tau^4}{45} - \frac{2\tau^6}{315} + O(\tau^8), \quad (22)$$

$$\int_0^1 M_{\rho_\tau}(s)^2 ds = \frac{2\tau^4}{945} + O(\tau^6). \quad (23)$$

Combining,

$$\mathcal{W}(\rho_\tau) = \left(\frac{\alpha}{45} + \frac{2\beta}{945} \right) \tau^4 + O(\tau^6), \quad \mathcal{W}(\rho_{\tau_*}) = O\left(\frac{d_{\text{head}}^4}{(L_{\text{eff}}^J)^2} \right). \quad (24)$$

The centered companion functional is $O(\tau^4)$. It is not the value difference of $\mathcal{C}_{\text{app}}$ itself and does not determine trained-model NLL.

A.7 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 8 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (25)$$

Proof. The Green’s function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (26)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (27)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (25). $\square$

An immediate corollary follows.

Corollary 9 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (28)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2 \sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity is an exact internal check on the two surrogate energy terms; it makes no trained-loss claim.

A.8 FROM SCALING STRUCTURE TO AN OPERATING DEFAULT

The post-softmax model uses $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ with $S_{\chi^2}(\tau) = \tau^4/(45d_{\text{head}}) + O(\tau^6)$ and $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)]$. Its leading stationary balance is

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}. \quad (29)$$

Thus the model supplies the powers of d_{head} and L ; the finite coefficient $c(\Pi) = \sqrt{45\lambda Q_1}$ depends on the chosen utility, base, and protocol. Setting $c=1$ is the deployed zero-search convention, not an independently derived constant.

Here

$$q(x) = \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x} \right)^2 = \text{Var}_{t \sim U[0,1]}[\cos(xt)]$$

and $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$. This is a specified phase-variance utility. It should not be identified with the exact oscillatory collision objective or the trained-transformer loss.

The finite- τ evidence tests a configuration-independent zero-search prior rather than a per-configuration point optimum. In a staged 99-run study, the default beats the midpoint-discretised Geo baseline in 7/9 configuration means and 18/27 seed pairs. The evaluated neighbours use the discrete ratios $0.75\times$, $1.25\times$, or $1.5\times$ of the formula; the independent exact-range factorial tests $0.75\times$, $1.00\times$, and $1.25\times$. Their ordering varies across configurations. These are tested multipliers, not bounds on a continuous basin. The evidence supports $c=1$ as a useful zero-search prior relative to that Geo baseline; it does not identify a per-configuration point optimum.

A.9 SCALING ANALYSIS OF τ^* : FROM SURROGATE TO OPERATING POINT

Numerical characterization of surrogate coefficients. We fit $K_{\text{app}}(\phi_i, \phi_j) = \alpha \delta_{ij} / \Delta\phi + \beta \min(\phi_i, \phi_j)$ to the exact collision kernel $K(\phi_i, \phi_j) = \frac{1}{2L} \left[\frac{\sin((\omega_1 - \omega_2)L)}{\omega_1 - \omega_2} + \frac{\sin((\omega_1 + \omega_2)L)}{\omega_1 + \omega_2} \right]$ evaluated at $K = d_{\text{head}}/2$ uniformly spaced channel positions in $[0, 1]$. Sweeping across $L \in \{128, \dots, 4096\}$, $d_{\text{head}} \in \{32, 64, 128, 256\}$, and $b \in \{10K, 100K, 500K\}$ yields:

$$\alpha \approx \frac{1}{2K} = \frac{1}{d_{\text{head}}}, \quad \beta \sim L^{-0.22}, \quad O(1). \quad (30)$$

The diagonal coefficient α is set by the kernel's diagonal value ($K(\phi, \phi) \approx 1/2$ for most channels) times the channel spacing $\Delta\phi = 1/K$. Crucially, α depends on d_{head} through the discretization ($K = d_{\text{head}}/2$) and is nearly independent of L . The off-diagonal coefficient β has weak power-law decay with L and negligible d_{head} dependence.

The two parameters called τ . Substituting into $\tau_{\text{surr}} = \sqrt{\beta/\alpha}$:

$$\tau_{\text{surr}} = \sqrt{\beta \cdot d_{\text{head}}} \sim \sqrt{d_{\text{head}}} \cdot L^{-0.11}. \quad (31)$$

This fitted-ratio scaling is neither the deployed rule nor evidence for it. The broadband surrogate selects the Cosh family; the separate post-softmax model in Proposition 10 balances an $O(\tau^2)$ utility with an $O(\tau^4)$ stiffness and yields

$$\tau_*^2 \propto \frac{d_{\text{eff}}^2}{L}. \quad (32)$$

The equality $c=1$ in the deployed rule is then evaluated by trained-model controls as described in §A.8; it is not obtained from the fitted β/α ratio.

A.10 POST-SOFTMAX OPERATING-POINT MODEL

The deployed scale is selected by a model separate from the surrogate that selects the shape. We state it here rather than in the main text because its small- τ expansion does not certify the deployed finite τ .

Proposition 10 (Semi-analytic operating-point scaling). *On the pure-tether family, assuming $Q_1(L, b) > 0$ and a diffuse post-softmax baseline $p_0=1/L$, the Pearson- χ^2 channel-load stiffness and the post-softmax transport utility admit small- τ expansions $S_{\chi^2}(\tau) = c_S \tau^4 / d_{\text{head}} + O(\tau^6)$ with $c_S = \frac{1}{45}$, and $U(\tau, L) = (d_{\text{head}}/L)[Q_0 + \tau^2 Q_1(L, b) + O(\tau^4)]$, where $Q_1(L, b) = \int_0^1 \eta(\phi) q(Lb^{-\phi}) d\phi$, $\eta(\phi) = (1-\phi)^2/2 - 1/6$ and $q(x) = 1/2 + \sin(2x)/(4x) - (\sin x/x)^2$. The leading-order stationary point of $\mathcal{F}(\tau) = \frac{1}{2} S_{\chi^2}(\tau) - \lambda U(\tau, L)$ satisfies*

$$\tau_*^2 = 45\lambda Q_1(L, b) \cdot \frac{d_{\text{head}}^2}{L}, \quad \tau_* \propto \frac{d_{\text{head}}}{\sqrt{L}}.$$

The explicit $d_{\text{head}}/\sqrt{L}$ factor is modulated by $Q_1(L, b)$, so this is a scaling structure rather than a pure power law, and the prefactor $c(\Pi) = \sqrt{45\lambda Q_1(L, b)}$ remains convention- and protocol-dependent. Two caveats bound what the proposition supports. The expansion is a small- τ statement, whereas the deployed configurations use τ near 4; at those values the exact cosh warp is evaluated directly and the series only motivates the functional form. The diffuse baseline $p_0=1/L$ is also a modelling assumption: trained attention is neither uniform nor stationary across heads, and the softmax-Jacobian effective window L_{eff}^J of §A.12 is the object that would replace it, but we do not measure it here. The deployed $c=1$ is therefore an empirical operating default, evaluated by the trained-model controls of §4.4, not a consequence of this proposition.

A.11 STIFFNESS FUNCTIONAL DERIVATION

The softmax transport objective in Proposition 10 requires choosing a stiffness functional $S(\tau)$ measuring the cost of departing from uniform density. We test a one-parameter family $S_p(\tau) = \frac{1}{d_{\text{head}}} \int_0^1 (\rho_\tau - 1)^2 / \rho_\tau^p d\phi$ using exact (non-Taylor) integrals of the normalized cosh density $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$.

Table 3: L -exponent γ (defined by $\tau_* \propto L^{-\gamma}$) under different stiffness functionals, computed via exact numerical optimization over $L \in [128, 4096]$. Target: $\gamma = 0.500$.

Stiffness	p	γ	Note
L^2 norm $\int (\rho-1)^2$	0	0.626	Original choice
Geometric mean	0.5	0.561	
Exponent-matched diagnostic	0.80	0.498	Sensitivity analysis, not used in deployment
Pearson χ^2	1.0	0.465	Channel-load variance penalty (axioms)
KL divergence	—	0.710	Worse than L^2
Reverse KL	—	0.581	
Jeffreys	—	0.621	$\approx L^2$
Rényi ($\alpha=2$)	—	0.807	

Why χ^2 . Two independent arguments favor $p=1$. (1) *Attention-load asymmetry*: each frequency channel at density $\rho(\phi)$ bears per-channel attention load $\propto 1/\rho$; weighting the squared departure by this load yields $\int (\rho-1)^2/\rho = \chi^2$. (2) *Discrete-grid effect*: the continuous L^2 integral, evaluated on the non-uniform EVQ-COSH grid, naturally acquires a $1/\rho$ Jacobian factor, producing discrete χ^2 .

Closed form. For $p=1$, the stiffness admits

$$S_{\chi^2}(\tau) = \frac{1}{d_{\text{head}}} \left[\frac{\sinh \tau \cdot \arctan(\sinh \tau)}{\tau^2} - 1 \right], \quad (33)$$

verified numerically to relative error $< 7 \times 10^{-6}$.

Exponent-matched diagnostic. As a sensitivity analysis, one can ask which member of the S_p family would match the target exponent $\tau_* \propto 1/\sqrt{L}$ exactly over this grid. The resulting $S_{\text{derived}}(\tau)$ lies near $p \approx 0.85$ (high-resolution sweep; consistent with 0.80 from the coarser grid), close to but distinct from the Pearson χ^2 choice $p=1$. We do not use this as a derivation of the stiffness; it is reported as sensitivity analysis. The full sweep is reproduced by `scripts/analysis/verify_stiffness_and_regime.py` (pure NumPy, ~ 60 s).

A.12 EFFECTIVE WINDOW L_{eff}^J VIA SOFTMAX-JACOBIAN RAYLEIGH QUOTIENT

From density perturbation to RoPE frequency shift. The EVQ-COSH pure-tether CDF is $F_\tau(\phi) = 1 - \sinh(\tau(1-\phi))/\sinh \tau$, with inverse-CDF warp $\Phi_\tau(u) = 1 - \tau^{-1} \text{arsinh}((1-u) \sinh \tau)$. Setting $\theta = \tau^2$ and expanding $\rho_\tau(\phi) = 1 + \theta a(\phi) + O(\theta^2)$ with $a(\phi) = (1-\phi)^2/2 - 1/6$, the quantile displacement takes the closed form

$$\Phi_\tau(u) = u + \tau^2 h(u) + O(\tau^4), \quad h(u) = -\int_0^u a(v) dv = -\frac{1}{6}u(1-u)(2-u), \quad (34)$$

so RoPE frequencies perturb as $\omega_k(\tau) = \omega_k^{(0)}[1 - \tau^2(\log b) h(u_k) + O(\tau^4)]$ with $\omega_k^{(0)} = b^{-u_k}$. Hence $\dot{\omega}_k := \partial_\theta \omega_k|_{\tau=0} = -(\log b) \omega_k^{(0)} h(u_k)$.

Explicit RoPE logit transport direction. Fix a layer/head and query i ; write each 2D RoPE pair as $q_{i,k}^{\mathbb{C}} = q_{i,2k-1} + i q_{i,2k}$, $k_{j,k}^{\mathbb{C}} = k_{j,2k-1} + i k_{j,2k}$, and $\alpha_{ij,k} = q_{i,k}^{\mathbb{C}} \overline{k_{j,k}^{\mathbb{C}}} = C_{ij,k} + i D_{ij,k}$. With relative position r_{ij} and attention scale s_{att} , the RoPE logit is $z_{ij}^{\text{rope}}(\tau) = s_{\text{att}}^{-1} \text{Re} \sum_k \alpha_{ij,k} \exp(ir_{ij} \omega_k(\tau))$. The leading logit transport direction $g_i(j) := \partial_\theta z_{ij}^{\text{rope}}|_{\tau=0}$ has the closed form

$$g_i(j) = \frac{(\log b) r_{ij}}{s_{\text{att}}} \sum_{k=1}^K \omega_k^{(0)} h(u_k) \left[C_{ij,k} \sin(r_{ij} \omega_k^{(0)}) + D_{ij,k} \cos(r_{ij} \omega_k^{(0)}) \right]. \quad (35)$$

This is the Fréchet derivative of the logit with respect to the EVQ-COSH allocation perturbation at fixed activations. A finite- τ implementation that avoids first-order phase expansion at large $r_{ij} \omega$ is

$$g_i^{(\tau)}(j) = \frac{1}{s_{\text{att}} \tau^2} \text{Re} \sum_{k=1}^K \alpha_{ij,k} \left[\exp(ir_{ij} b^{-\Phi_\tau(u_k)}) - \exp(ir_{ij} b^{-u_k}) \right], \quad (36)$$

with $g_i(j) = \lim_{\tau \rightarrow 0} g_i^{(\tau)}(j)$.

Primary definition: Rayleigh quotient of the softmax Jacobian. For the trained attention distribution $p_i(j)$ on keys $j \in \mathcal{K}_i$, $J_{\text{sm}}(p_i) = \text{diag}(p_i) - p_i p_i^\top$ satisfies $g_i^\top J_{\text{sm}}(p_i) g_i = \text{Var}_{j \sim p_i}[g_i(j)] = \frac{1}{2} \sum_{j,k} p_i(j) p_i(k) (g_i(j) - g_i(k))^2$. Since $J_{\text{sm}}(p_i) \mathbf{1} = 0$, we center g_i by the key-wise mean operator P_i . The effective length is then defined by the *ratio-of-means*

$$\frac{1}{L_{\text{eff}}^J} = \kappa_{\text{att}} = \frac{\mathbb{E}_{\ell,h,i}[g_i^\top J_{\text{sm}}(p_i) g_i]}{\mathbb{E}_{\ell,h,i}[\|P_i g_i\|_2^2]}, \quad (37)$$

This is the directional curvature of the *schedule-change KL*: for $p_\epsilon = \text{softmax}(z + \epsilon g)$, $D_{\text{KL}}(p \| p_\epsilon) = \frac{1}{2} \epsilon^2 g^\top J_{\text{sm}}(p) g + O(\epsilon^3)$, and for diffuse $p=1/L$ the quotient reduces to $1/L$. The phase-variance utility Q_1 in Proposition 10 is a distinct pairwise positional functional. Replacing its $1/L$ factor by $1/L_{\text{eff}}^J$ is therefore a possible attention-conditioned refinement rather than a consequence of the KL identity. The empirical claims in this paper use the registered length-based operating rule.

A.13 PURE-TETHER BRANCH: FISHER-FORCING COEFFICIENT AND RESIDUAL BOUND

Setup. The full stationary ODE under $\mathcal{C}_{\text{app}}$ plus a Fisher utility is $\rho'' - \tau^2 \rho = \gamma b^{-2\phi}$. Writing $s = 2 \log b$ and $q_s(\phi) = e^{-s\phi}$, the pure-tether branch $\rho_{\text{EVQ}}(\phi) = \tau \cosh(\tau(1-\phi)) / \sinh \tau$ solves the homogeneous problem; the full solution is $\rho_{\text{full}} = \rho_{\text{EVQ}} + \delta \rho_F$.

Explicit form of $\gamma(b)$. The Fisher utility in the Euler equation $\alpha \rho + \beta(G\rho) - \lambda_F V_b + \nu = 0$ has $V_b(\phi) = \eta_F(\phi) b^{-2\phi}$, where $\eta_F(\phi)$ is the activation-conditioned Fisher coefficient with

$$\eta_F(\phi_k) = \frac{1}{2 s_{\text{att}}^2} \mathbb{E}_{\ell,h,i,j}[w_{\ell hij} |\alpha_{\ell hij,k}|^2], \quad \alpha_{\ell hij,k} = q_{i,k}^{\mathbb{C}} \bar{k}_{j,k}^{\mathbb{C}}. \quad (38)$$

Using $(G\rho)'' = -\rho$, differentiating the Euler equation twice gives $\rho'' - \tau^2 \rho = (\lambda_F/\alpha) V_b''$. For slowly varying η_F compared with $e^{-s\phi}$,

$$\gamma(b) = \frac{\lambda_F}{\alpha} \bar{\eta}_F s^2 \left[1 + O\left(\frac{|\eta_F'|}{s \bar{\eta}_F} + \frac{|\eta_F''|}{s^2 \bar{\eta}_F}\right) \right], \quad s = 2 \log b. \quad (39)$$

The full $\eta_F(\phi)$ -local version is $\gamma_{\text{loc}}(\phi; b) = (\lambda_F/\alpha)[\eta_F''(\phi) - 2s\eta_F'(\phi) + s^2\eta_F(\phi)]$.

Suppression is L^1/CDF , not pointwise. The particular solution $\delta \rho_F^{\text{part}} = \gamma(b) q_s / (s^2 - \tau^2)$ combined with (39) gives pointwise amplitude

$$\delta \rho_F^{\text{part}}(\phi) \approx \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{s^2 - \tau^2} e^{-s\phi}, \quad (40)$$

so the $(s^2 - \tau^2)^{-1}$ factor is cancelled by the s^2 in $\gamma(b)$, and $\delta \rho_F^{\text{part}}$ is *not* pointwise small in s . The true suppression comes from the boundary-layer width of q_s : $\|q_s\|_1 = (1 - e^{-s})/s = O(1/\log b)$ and $\|q_s\|_2 = O(1/\sqrt{\log b})$. Let $C_{\text{bc}}^{(p)}(\tau)$ denote the finite L^p operator norm of the homogeneous correction enforcing the boundary and mass constraints, away from resonance. Then

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_1 \leq C_{\text{bc}}^{(1)}(\tau) \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{|s^2 - \tau^2|} \frac{1 - e^{-s}}{s} = O\left(\frac{1}{\log b}\right), \quad (41)$$

$$\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2 \leq C_{\text{bc}}^{(2)}(\tau) \frac{\lambda_F \bar{\eta}_F}{\alpha} \frac{s^2}{|s^2 - \tau^2|} \sqrt{\frac{1 - e^{-2s}}{2s}} = O\left(\frac{1}{\sqrt{\log b}}\right), \quad (42)$$

and the CDF-level error is $\|F_{\text{full}} - F_{\text{EVQ}}\|_\infty \leq \|\delta \rho_F\|_1 = O(1/\log b)$. For fixed Fisher coefficient and away from resonance, forcing is therefore small in CDF / L^1 transport at large b , but not necessarily pointwise small near $\phi=0$.

Warp amplification at large τ . Via the inverse-CDF map,

$$\|\phi_{\text{full}} - \phi_{\text{EVQ}}\|_\infty \leq \frac{\|\delta \rho_F\|_1}{\inf_\phi \rho_{\text{full}}(\phi)}, \quad \inf_\phi \rho_{\text{EVQ}}(\phi) = \frac{\tau}{\sinh \tau}, \quad (43)$$

so, when the residual is small enough that ρ_{full} remains positive and comparable to ρ_{EVQ} , CDF perturbations can be amplified at the scale $\sinh \tau / \tau$. The bound does not justify omitting the forcing branch at large τ .

Dimensionless diagnostic. Using $\|\rho_{\text{EVQ}} - 1\|_2 = \tau^2/\sqrt{45} + O(\tau^4)$ and $\Gamma_b := (\lambda_F \bar{\eta}_F / \alpha) s^2 / (|s^2 - \tau^2| \tau^2)$, the L^2 -level ratio is

$$R_F := \frac{\|\rho_{\text{full}} - \rho_{\text{EVQ}}\|_2}{\|\rho_{\text{EVQ}} - 1\|_2} \leq C_{\text{bc}}^{(2)} \Gamma_b \sqrt{45} \sqrt{\frac{1-e^{-2s}}{2s}} = O\left(\frac{1}{\sqrt{\log b}}\right). \quad (44)$$

Second-order utility deficit. Let A_τ be the constrained Hessian of the surrogate on zero-mass perturbations and let $q_s^\perp$ be the projection of q_s onto that tangent space. The full branch improves the quadratic objective by

$$\Delta_F = \frac{1}{2} \gamma(b)^2 \langle q_s^\perp, A_\tau^{-1} q_s^\perp \rangle, \quad (45)$$

so dropping the forcing term is $O(\gamma(b)^2)$: if $R_F \leq r$, the missed correction is $O(r^2)$ relative to the tether deformation energy.

A.14 CHANNEL-LOAD MOTIVATION FOR PEARSON χ^2

Normalization convention. Throughout this paper, $S_{\chi^2}(\tau)$ denotes the d_{head} -normalized stiffness $S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 (1-\rho_\tau)^2 / \rho_\tau d\phi$, with leading expansion $\tau^4/(45 d_{\text{head}}) + O(\tau^6)$ consistent with Proposition 10. Identities below that produce a bare $\tau^4/45$ leading coefficient apply to $d_{\text{head}} S_{\chi^2}(\tau)$ (the un-normalized integral); the d_{head} factor enters via the surrogate diagonal $\alpha=1/d_{\text{rot}}$ ($=1/d_{\text{head}}$ for MHA via §A.9; $=1/d_{\text{rot}}$ for compressed-RoPE MLA via §G.1).

Channel-load axioms. Inverse-CDF quantization places channels with cell width $\ell(\phi)=1/\rho(\phi)$ and samples channels according to $dP_\rho=\rho d\phi$. Under the modeling axioms (i) channel-measure averaging, (ii) load ℓ as the primitive local quantity, (iii) second central moment, (iv) uniform density has zero stiffness, (v) local additivity, the canonical load-variance penalty is

$$S(\rho) = \text{Var}_{\phi \sim \rho d\phi} \left[\frac{1}{\rho(\phi)} \right] = \int_0^1 \frac{(1-\rho(\phi))^2}{\rho(\phi)} d\phi = S_{\chi^2}(\rho). \quad (46)$$

We treat this as a canonical motivated choice under the stated axioms, not a uniqueness claim.

Load-moment exponent. Distinct from the weighted- L^2 family $S_p(\tau) = \frac{1}{d_{\text{head}}} \int (\rho_\tau - 1)^2 / \rho_\tau^p d\phi$ used in Table 3 (where Pearson χ^2 is $p=1$), we also examine a load-moment family $S_p^{\text{mom}}(\rho) = \mathbb{E}_\rho[(1/\rho-1)^p]$. Substituting the EVQ-COSH density and minimizing $S_p^{\text{mom}} - \lambda U$ gives $\tau^{2(p-1)} \asymp d_{\text{head}}^2 / L_{\text{eff}}^J$, hence $\tau^* \propto (L_{\text{eff}}^J)^{-1/(2p-2)}$; only $p=2$ yields the observed $L^{-1/2}$ exponent within this family. The two families parametrize different f -divergence neighborhoods and need not match on the finite- τ grid.

f -divergences at finite τ . Smooth f -divergences (KL, Jeffreys, Rényi) are locally quadratic around $\rho=1$. Over the finite grid used in Table 3, however, their fitted exponents range from 0.58 to 0.81. This is model sensitivity, not a trained-PPL comparison.

A.15 DISCRETE-CHANNEL TRANSPORT GAP AND THE MLA CHANNEL-SCARCITY MECHANISM

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. Direct $\|\rho_K - \rho\|$ is ill-posed: one cannot subtract an atomic measure from a density. We therefore work in Wasserstein distance for transport and in a quantile-cell histogram ρ_K for density.

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (47)$$

Defining the quantile-cell histogram $\rho_K(\phi)=1/(K|I_k|)$ on $I_k=[Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (48)$$

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$, $m_\tau = \tau/\sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (49)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km} + O(K^{-2}). \quad (50)$$

The singular δ -component of K_{app} must be interpreted as a cell average (not as a literal atom-kernel pairing), after which the bound applies to the smoothed kernel.

High-resolution distortion. For a smooth weighted distortion w ,

$$\mathcal{D}_K[\rho] \approx \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (51)$$

This is the finite-channel mechanism: a shaped ρ can reduce the weighted inverse-density load $\int w/\rho^2$; the absolute benefit scales as K^{-2} for smooth quantization error and as K^{-1} for transport/kernel terms.

Channel scarcity amplifies the density-shape correction. The standard MHA configuration has $d_{\text{head}}=d_{\text{rope}}=64$ and $K=32$; the MLA study has $d_{\text{head}}=64$, $d_{\text{rope}}=32$, $d_{\text{nope}}=32$ and $K=16$. Halving K scales the K^{-2} component of the distortion bound by $4\times$ and the K^{-1} component by $2\times$. The observed MLA gain (§G.1) is a test of the predicted direction, not a controlled quantitative test of either bound.

Caveat. At large τ , $m_\tau=\tau/\sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds can therefore become vacuous in the same regime where the Fisher-residual analysis no longer guarantees a small warp error (§A.13).

B EXPERIMENTAL DETAILS

B.1 SCOPE OF THIS APPENDIX

The exact-range and mature-model protocols are documented separately in Apps. C and F. This section records the submitted scratch-model and cross-modal experiments that remain supporting evidence for the substrate-range and cross-modal scope claims. The learned-frequency comparator appears in App. E.

B.2 REPRODUCIBILITY SNAPSHOT

Table 4: Compact reproducibility summary for supporting scratch-model experiments.

Setting	Train length	Scale	Seeds	Main datasets
EVQ-COSH ($\tau=1.414$) $\times$ YARN-STYLE	2048	454M	3 + 3	FineWeb-Edu + 10% passkey mix
MLA	8192	432M	3	FineWeb-Edu

Table 5 lists the training hyperparameters shared across the supporting text experiments.

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base $b=500K$ unless stated otherwise.

Table 5: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$, $\epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 3×10^{-4} (125M), 2×10^{-4} (454M/MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 16 (125M), 2–4 (454M/750M), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

Table 6: 750M continued-pretraining check (seed 42; 2K $\rightarrow$ 4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials.

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Strict AR exact @8K	0%	77.5%
Global strict AR exact	66.67%	92.5%

B.3 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. The EVQ-COSH continuation uses midpoint quantisation with $\tau=1.5$. Both arms are full-parameter continuations from the shared checkpoint.

B.4 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau=K_t/\sqrt{T}=2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ (0.53 $\times$), which selects the supporting video configuration.

B.5 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

We evaluate a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames (4 $\times$) with temporal YARN-STYLE. For each seed, the Geo and EVQ-COSH ($\tau=1.5$) branches are trained in the same head-to-head run.

Table 7 reports two independent seeds. The within-seed contrasts favour EVQ-COSH on train, all-frame extrapolation, and far-frame extrapolation MSE in both runs; the two-seed mean deltas are -26% , -16% , and -33% , extending the scope check to bidirectional 3D RoPE.

Table 7: DiT head-to-head evaluation (129.6M, bidirectional attention, 3D RoPE). Each seed is one shared-run comparison; values are denoising MSE at $t=0.5$.

Seed	Metric (YARN-STYLE)	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	−21%
42	All extrap	0.00724	0.00606	−16%
42	Far extrap	0.00989	0.00639	−35%
137	Train MSE	0.01113	0.00761	−32%
137	All extrap	0.00713	0.00598	−16%
137	Far extrap	0.00856	0.00601	−30%
Mean Δ		−26% / −16% / −33%		

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. In the pre-specified diagnostic, $\tau=1.2$ leaves 11/16 channels above this threshold (effective rank 9), whereas $\tau=1.5$ leaves 12/16 (rank 10). Repeating the head-to-head comparison at base=1,000 places 13/16 geometric channels above the threshold. At seed 42, $\tau=1.2$ and $\tau=1.5$ differ by less than 0.4% on the reported metrics and both improve far-frame MSE by about 48% relative to Geo, supporting the same low-phase-channel diagnostic.

Table 8: Base = 1,000 DiT head-to-head (129.6M, seed 42, YARN-STYLE). The two EVQ-COSH settings are within 0.4% on the reported metrics in this run.

τ	Train MSE	All extrap	Far extrap	$\Delta_{\text{far vs Geo}}$
0.0 (Geo)	0.02037	0.01274	0.01646	—
1.2	0.00933	0.00820	0.00859	−48%
1.5	0.00930	0.00822	0.00858	−48%

B.6 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5 \times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 1 separately measures incompatibility after a frozen post-hoc swap.

C IDENTIFICATION PROTOCOL AND PROVENANCE

C.1 EXACT-RANGE CONTROL AT 151.9M

What is pinned. The control is designed so that every scalar quantity a base/range method can express is identical between arms. Within each seed pair the two arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2 = 30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6 \times 10^{-4}/6 \times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Baseline:* a paper-faithful reimplementation of the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026b), at training base 256. The baseline implements that published rule directly. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles q_k , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Concretely, let $q_k = \Phi_\tau((k + \frac{1}{2})/K)$, $s_k = (q_k - q_0)/(q_{K-1} - q_0)$, and $R = ((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R\frac{k}{K-1}\right). \quad (52)$$

Hence $k=0, K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed*: each checkpoint retains its training range. *target-matched*: the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Table 9: Exact-range anchored EVQ-COSH minus FMRoPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean±SD	Seed 42	Seed 137	Seed 256	Mean±SD
256	+0.033	+0.029	+0.017	+0.026 ± 0.008	+0.033	+0.029	+0.017	+0.026 ± 0.008
512	−0.478	−0.271	−0.094	−0.281 ± 0.192	+0.061	+0.053	+0.067	+0.060 ± 0.007
1,024	−0.205	−0.191	−0.132	−0.176 ± 0.039	+0.182	+0.156	+0.344	+0.227 ± 0.102
2,048	−0.113	−0.181	−0.143	−0.146 ± 0.034	+0.279	+0.399	+0.701	+0.460 ± 0.218

Under *fixed*, anchored EVQ-COSH is favoured by 3/3 seeds at every OOD length; under *target-matched*, FMRoPE is favoured by 3/3 seeds at every OOD length. The fixed-range sign agrees across all seeds and OOD lengths; the 1K/2K magnitudes are the most stable across seeds.

Reading. The *fixed* column identifies the optimisation variable: interior allocation is not reducible to scalar base/range selection. Retargeting support changes absolute phase coverage and therefore changes which interior allocation is preferred. This is the expected interaction between the two coordinates: target-specific support should be selected jointly with allocation, while the *fixed* column isolates allocation when one table serves multiple lengths (§4.1).

C.2 EXACT-RANGE FACTORIAL AT 50.9M

Design. The schedule arms, including the $1.25\times$ multiplier, were specified before any result was inspected. Grid: 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 structural configurations; 5 schedules (Geo, anchored EVQ-COSH at $0.75\times$, $1.00\times$ [the formula point], $1.25\times$, and a deformation-matched exponential); 3 seeds. Main factorial 180/180 completed, boundary- τ follow-up 12/12 completed, covering the complete pre-specified matrix.

Table 10: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times /4 \times /8 \times$, paired within configuration and seed. All arms were pre-specified; no contrast was registered as primary. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. C).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH $0.75 \times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH $1.25 \times$ – Geo	− 0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$, never an arithmetic mean of raw PPL.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75 \times$ 6.202858 (494.16); EVQ-COSH $1.00 \times$ 6.202094 (493.78); EVQ-COSH $1.25 \times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5 \times$ arm is better; at formula $\tau=8$ the $0.75 \times$ arm is. Optima move inward from the two tested extremes; these discrete arms do not locate continuous basin boundaries or yield a replacement closed-form rule.

Scope. This is a short-budget 50.9M mechanistic NLL control trained on a deterministically repeated WikiText-2 raw stream, a different corpus from the 151.9M control and from the mature-scale runs. The factorial establishes cross-configuration *direction*, while the 151.9M control provides the effect size within its configuration. Scale is evaluated in §4.2.

D THE YARN-STYLE OPERATOR

Every arm marked +YARN-STYLE uses the repository’s fixed-index operator, inspired by the NTK-by-parts/YaRN family (Peng et al., 2024): high-frequency channels are left nearly unchanged and lower-frequency channels receive progressively more range scaling. It is held *identical* across the arms being compared — same implementation, same fixed cut indices, same scale s — so that the only difference between Geo + YARN-STYLE and EVQ-COSH + YARN-STYLE is the training-time table each substrate presents to the same operator. It therefore measures substrate-dependent leverage at the fixed tested scale.

Implemented components. For K frequency channels, the scaler uses fixed channel-index cut points $k_{\text{lo}} = \lfloor 0.20K \rfloor$ and $k_{\text{hi}} = \lfloor 0.90K \rfloor$. Between them it applies the cubic smoothstep $r(x) = x^2(3 - 2x)$ and replaces each inverse frequency ω_k by

$$\omega_k / \left(s^{r_k} t^{r_k/2} \right), \quad t = 1 + 0.07 \log_2 s.$$

Its boundaries are fixed by channel index; the factor t is folded into frequency scaling. These implementation choices define YARN-STYLE and distinguish it from the cited YaRN reference method.

E LEARNED-FREQUENCY COMPARATOR

Table 11: Learned-frequency comparator (125M, $L_{\text{train}}=128$, FineWeb-Edu, $128 \rightarrow 8K$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). Allocation-shape attribution uses the fixed, zero-parameter analytic schedules of §4.1; this separately reported arm is a layer-shared learnable inverse-frequency baseline — one learned scalar per band, shared across layers and heads — and therefore belongs to the learned-*table* family (Karypis et al., 2026); DAPE belongs to the learned positional-*operator* family (Zheng et al., 2024).

Method (128 $\rightarrow$ 8K)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 ± 1.3	437.9 ± 12.2	−14.8%
Learned inv-freq (layer-shared)	32	183.6	455.3	−11.4%
same, $10\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm in Table 11 is a layer-shared learnable inverse-frequency baseline: one learned scalar per frequency band, shared across layers and heads. It received a dedicated positional-parameter learning-rate sweep at $10\times$ and $100\times$ the base rate, and the stronger $100\times$ setting is the 455.3 row. It answers the learned-table tuning question. Allocation-shape attribution uses the fixed analytic schedules under an identical zero-parameter operator in §4.1.

On method identity: DAPE (Zheng et al., 2024) is a learned positional *operator*, whereas this comparator learns inverse frequencies directly and belongs to the learned-*table* family later studied at scale by Karypis et al. (2026). The distinction matters for what the row can be read as evidence for. Read as a learned-table comparison, its direction is consistent with the independent finding that learned tables degrade more sharply than standard RoPE under naive extrapolation.

F MATURE-SCALE PROTOCOLS

F.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use endpoint quantisation with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (OLMo Team, 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical

Table 12: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA. Training and evaluation rows are disjoint and share the 13 generator families, measuring task-family-adapted length transfer.

Model / protocol	Arm / metric	$1\times$	$2\times$	$4\times$
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

tokens, but explicit position IDs expose target-range relative phases. This is therefore a physical-memory cap, not phase-free 4K training. On 8K `niah_single_1`, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set. Natural-text NLL and the distinct autoregressive endpoints are kept separate in the discussion below and in Table 12.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start independently from the matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O LoRA tensor (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length, Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 12 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals condition on the single trained pair and quantify evaluation-row uncertainty, not training-seed variability.

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64 at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is $+0.0724/+0.0381/-0.0437/-0.1351$ at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K. This is one same-initialisation, same-scientific-recipe natural-text LM comparison; capability is evaluated by the separate autoregressive protocols above.

F.2 LLAMA-3-8B-INSTRUCT

The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$; Native retains the pretrained model’s original frequency table. The training objectives, adapters, and endpoints remain separate below.

Probability and causal source use. In the matched seed-42, 300-step Q/K/V/O LoRA protocol, EVQ-COSH-minus-Native NLL is $+0.390/-1.510/-2.048$ at 8/16/32K. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Removing the remote gold block from every head changes NLL by -0.0095 for Native and $+1.5055$ for EVQ-COSH. These are matched teacher-forced probability and causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

RULER-family continuation. A separate 516-step continuation uses identical physical-8K supervision over all 13 RULER families, with one seed per arm and disjoint generated training/evaluation rows. Evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 12 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

Endpoint scope. NLL, attention routing, strict autoregressive exact match, multi-hop QA, and RULER remain separate endpoints. The 2Wiki and RULER continuations use the same task families at train and test with disjoint rows, establishing task-family-adapted length transfer. The natural-text and routing rows evaluate probability and causal source use.

G SUPPORTING RESULTS

G.1 MULTI-HEAD LATENT ATTENTION

We test the finite-budget hypothesis in a 432M-parameter MLA transformer with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified $\tau=1.414$ setting. This experiment tests frequency allocation at fixed architecture.

Table 13: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator, which is distinct from YARN-STYLE.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ-COSH reduces 16K PPL by 31.1% (138.8 $\rightarrow$ 95.6), while 8K PPL changes by +1.1%. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond. These rows establish the tested ordering for this architecture and protocol.

G.2 COMPOSITION WITH YARN-STYLE

H SUPPORTING EXPERIMENTS (FROM MAIN TEXT)

The following results record additional submitted protocols with their seed scope.

Cross-modal scope. In two 129.6M DiT shared-run comparisons, the EVQ-COSH branch has lower denoising MSE on train, all extrapolation frames, and far extrapolation frames in both seeds. Full values and the one-seed base diagnostic are in §B.5.

Table 14: **Substrate $\times$ range composition** (454M, $L_{\text{train}}=2048$, 10% passkey mix, 3 seeds). YARN-STYLE is the repository’s fixed-index operator (App. D), applied at the *same* fixed scale $s=8$ to both substrates. PK is teacher-forced NLL-gap retrieval; PK@8K is mean $\pm$ std, other entries are 3-seed means.

Method	PK@8K	PK@12K	PK@16K	PPL@8K	PPL@16K
Geo	41 $\pm$ 5%	57%	51%	161.9	253.2
Geo+YARN-STYLE	61 $\pm$ 3%	59%	51%	82.9	157.7
EVQ-COSH	53 $\pm$ 8%	63%	50%	150.3	229.5
EVQ-COSH+YARN-STYLE	100$\pm$0%	79%	68%	70.9	107.5

Table 15: $L=256$ YARN-STYLE composition study (125M, 3-seed mean PPL). Among the tested settings, $\tau=4.0$ has the lowest mean PPL at every reported extrapolation ratio.

Method ($L=256$)	$4\times$	$8\times$	$16\times$	$32\times$
Geo+YARN-STYLE	60.7	97.5	190.4	260.2
EVQ-COSH ($\tau=2$)+YARN-STYLE	51.6	66.7	107.9	141.7
EVQ-COSH ($\tau=4$)+YARN-STYLE	50.9	60.9	84.1	99.6